DEPI Graduation Project

Test Plan

Team Code	CAI5_SWD6_G1_Team1
Team Members	Hagar Alaaeldien (Team Lead), Rokaia Ahmed, Esraa Mohamed, Nada Abou Elfadl
Sprint duration	1 week
Selected systems	SauceDemo – Manual & UI Automation | Fake Store API – API Testing
Component	**Details**
Objectives	The goal is to establish a structured testing process for the selected web application and REST API, ensuring that their core functionality is validated through manual, automated UI and API testing. **Specific objectives include:** • Verify the core user workflows of SauceDemo through manual and automated testing. • Validate the available Fake Store API endpoints, response codes, response payloads and error handling. • Apply Selenium with Java for UI automation and Postman for API testing.
Approach	The testing approach combines manual, automated UI and API testing. • **Manual testing:** SauceDemo will be tested through functional and exploratory testing of its main user workflows. • **UI automation:** Selenium WebDriver with Java will automate selected SauceDemo workflows. Page Object Model (POM) will be used to support a maintainable framework. • **API testing:** Postman will be used to test Fake Store API requests, response codes, payloads and error handling. • **Test management:** Jira will be used to track test activities, defects and sprint progress. • **Version control:** GitHub will be used to manage automation code and project documentation. • **BDD, performance and CI:** BDD feature files will describe selected automated scenarios; Apache JMeter will be used for light-load performance testing; and Jenkins will run the automated tests as part of the CI pipeline (Sprints 2–4). • **Entry criteria:** Testing starts when the test plan is reviewed, the required tools are installed, both systems are reachable, and the test cases are reviewed and logged in Jira. • **Exit criteria:** Testing is complete when all planned test cases are executed, every failure is logged as a defect in Jira, no critical defects remain open, and the test execution report is delivered.
Scope	**In Scope:** • SauceDemo – login and logout. • SauceDemo – product inventory, product details and sorting. • SauceDemo – adding/removing products from the cart and cart validation. • SauceDemo – checkout information, order overview and order completion. • Fake Store API – available e-commerce REST API endpoints and their supported operations. • API validation – request/response behavior, status codes, response payloads and error handling. • Manual functional and exploratory testing, selected Selenium UI automation scenarios, and Postman API testing. **Out of Scope:** • Testing or modifying the internal source code, infrastructure or databases of the public demo systems. • Features or API operations that are unavailable or not documented by the selected systems. • Real payments, real customer information or production transactions.
Test Deliverables	• Test Plan Document. • Project Roadmap covering all four sprints. • Manual and automated UI test cases for SauceDemo. • Postman API collection and API test results for Fake Store API. • Defect reports and test execution reports. • Final test summary and project documentation.
Dependencies	• Access to SauceDemo: https://www.saucedemo.com • Access to Fake Store API and its documentation: https://fakestoreapi.com • Availability and stability of both public demo systems. • Java/JDK, Selenium WebDriver, a compatible browser/driver, Postman, Jira and GitHub. • Team access to computers, stable internet and a shared communication channel. **Contingency plan:** • If either public service is unavailable, document the issue and retry later. Use documented examples or mock responses only for practice/planning, and clearly distinguish them from live API test results. • If API access rules or available endpoints change, review the current documentation and update the planned tests.
Test Environment	• Hardware: Team members' laptops/desktops with internet access. • Operating system: Microsoft Windows. • Browser: Current stable Google Chrome for SauceDemo UI testing. • UI automation: Java/JDK, Selenium WebDriver and compatible browser driver.

	• API testing: Postman. • Test management: Jira. • Version control: Git and GitHub. • Performance testing: Apache JMeter. • Network: Stable internet connection to access SauceDemo and Fake Store API.
Risk Management	**Identified Risks:** • SauceDemo or Fake Store API may become unavailable or unstable. • API endpoints, access requirements or response behavior may change. • Unequal task distribution or scheduling conflicts may affect sprint progress. • Differences in Java, browser or driver configurations may cause automation setup issues. • The one-week sprint duration may limit the time available for planning and review. **Mitigation Strategies:** • Check system availability and API documentation before testing; record service outages and retry later. • Keep API requests and test assumptions aligned with the current Fake Store API documentation. • Distribute tasks equally and ensure every member participates in all three testing types. • Document setup requirements and use a consistent team workflow through GitHub.
Schedule	**Timeline – four one-week sprints:** • **Sprint 1 – Planning:** Select projects; define test objectives, scope, approach, schedule, entry/exit criteria, resources and risks. Deliverables: Test Plan and Project Roadmap. • **Sprint 2 – Analysis & Design:** Perform exploratory testing; develop manual, automation and API test cases; select automation candidates; prepare initial framework setup and first BDD feature file; log test cases in Jira. Deliverables: initial manual/automated test cases, Jira test records and first BDD feature file. • **Sprint 3 – Implementation:** Set up GitHub workflow; create and validate Postman API scripts; configure JMeter; develop the automation framework. Deliverables: GitHub repository/framework, Postman collection and results, JMeter scripts. • **Sprint 4 – Test Completion & Execution:** Optimize the framework; execute UI, API and performance tests; analyze and report results; integrate Jenkins CI; finalize documentation. Deliverables: optimized framework, full execution report, final documentation, CI/CD pipeline and Test Completion Report.
Roles and Responsibilities	All four members will participate in manual testing, Selenium automation and API testing. Work will be divided equally, with each member responsible for approximately 25% of the planned work in each testing stream. • **Hagar Alaaeldien – Team Lead:** Contributes an equal share to SauceDemo manual testing, Selenium automation and Fake Store API testing; coordinates sprint planning, tracks progress and consolidates team reporting. • **Rokaia Ahmed:** Completes an equal share of SauceDemo manual testing, Selenium automation and Fake Store API testing; participates in reviews and reporting. • **Esraa Mohamed:** Completes an equal share of SauceDemo manual testing, Selenium automation and Fake Store API testing; participates in reviews and reporting. • **Nada Abou Elfadl:** Completes an equal share of SauceDemo manual testing, Selenium automation and Fake Store API testing; participates in reviews and reporting. The team will rotate task ownership across sprints to ensure that no member is assigned exclusively to one testing type. Jira will be used for task/defect tracking, GitHub for code collaboration, and the agreed team communication channel for stand-ups and decisions.